

 Add Fortune on Google for similar content.

Add us on Google

AI • PRODUCTIVITY

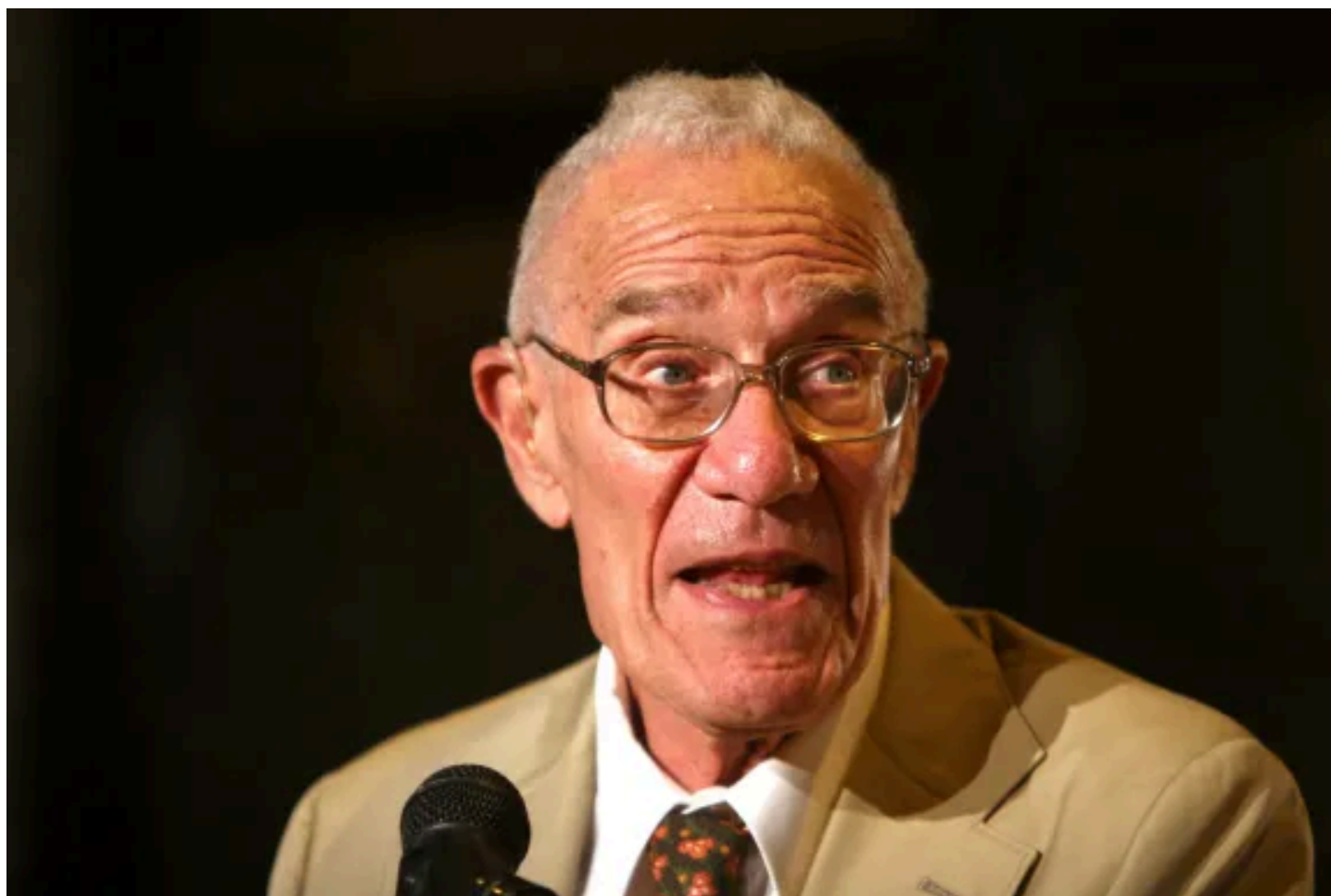
Thousands of CEOs admit AI had no impact on employment or productivity—and it has economists resurrecting a paradox from 40 years ago

By **Sasha Rogelberg**

Reporter



April 19, 2026, 10:58 AM ET



Nobel laureate and economist Robert Solow noticed a productivity paradox in the IT age of the 1980s that economists today see reflected in the AI boom.

LIOR MIZRAHI—GETTY IMAGES

In 1987, economist and Nobel laureate Robert Solow made a stark observation about the stalling evolution of the Information Age: Following the advent of transistors, microprocessors, integrated circuits, and memory chips of the 1960s, economists and companies expected these new technologies to disrupt workplaces and result in a surge of productivity. Instead, [productivity growth slowed](#), dropping from 2.9% from 1948 to 1973, to 1.1% after 1973.

Recommended Video

Newfangled computers were actually at times [producing too much information](#), generating agonizingly detailed reports and printing them on reams of paper. What had promised to be a boom to workplace productivity was for several years a bust. This unexpected outcome became known as Solow's productivity paradox, thanks to the economist's observation of the phenomenon.

"You can see the computer age everywhere but in the productivity statistics," Solow wrote in a [New York Times Book Review article](#) in 1987.

Data on how C-suite executives are—or aren't—using AI shows history is repeating itself, complicating the similar promises economists and Big Tech founders made about the technology's impact on the workplace and economy. Despite 374 companies in the S&P 500 mentioning AI in earnings calls—most of which said the technology's implementation in the firm was entirely positive—according to a [Financial Times analysis](#) from September 2024 to 2025, those positive adoptions aren't being reflected in broader productivity gains.

A [study](#) published in February by the National Bureau of Economic Research found that among 6,000 CEOs, chief financial officers, and other executives from firms who responded to various business outlook surveys in the U.S., U.K., Germany, and Australia, the vast majority see little impact from AI on their operations. While about two-thirds of executives reported using AI, that usage amounted to only about 1.5 hours per week, and 25% of respondents reported not using AI in the workplace at all. Nearly 90% of firms said AI has had no impact on employment or productivity over the past three years, the research noted.

However, firms' expectations of AI's workplace and economic impact remained substantial: Executives also forecast AI will increase productivity by 1.4% and increase output by 0.8% over the next three years. While firms expected a 0.7% cut to employment over this time period, individual employees also surveyed saw a 0.5% increase in employment.

Is AI actually making people more productive?

In 2023, MIT researchers claimed AI implementation could [increase a worker's performance by nearly 40%](#) compared to workers who didn't use the technology. But emerging data failing to show these promised productivity gains has led economists to wonder when—or if—AI will offer a return on corporate investments, which [swelled to more than \\$250 billion](#) in 2024.

“AI is everywhere except in the incoming macroeconomic data,” Apollo chief economist Torsten Slok wrote in a [blog post](#), invoking Solow’s observation from nearly 40 years ago. “Today, you don’t see AI in the employment data, productivity data, or inflation data.”

Slok added that outside of the Magnificent Seven, there are “no signs of AI in [profit margins](#) or [earnings expectations](#).”

Slok cited a slew of academic studies on AI and productivity, painting a contradictory picture about the utility of the technology. Last November, the Federal Reserve Bank of St. Louis published in its [State of Generative AI Adoption report](#) that it observed a 1.9% increase in excess cumulative productivity growth since the late-2022 introduction of ChatGPT. A [2024 MIT study](#), however, found a more modest 0.5% increase in productivity over the next decade.



PAID CONTENT

Crypto markets are entering a new phase of maturity [↗](#)

FROM BINANCE

“I don’t think we should belittle 0.5% in 10 years. That’s better than zero,” study author and Nobel laureate Daron Acemoglu said at the time. “But it’s just disappointing relative to the promises that people in the industry and in tech journalism are making.”

Other emerging research can offer reasons why: Workforce solutions firm [ManpowerGroup’s 2026 Global Talent Barometer](#) found that across nearly 14,000

workers in 19 countries, workers' regular AI use increased 13% in 2025, but confidence in the technology's utility plummeted 18%, indicating persistent distrust.

AI adoption can even be counterproductive at a certain point, according to a [study](#) conducted by [Boston Consulting Group](#), leading to "AI brain fry." In a survey of 1,488 full-time U.S.-based workers, respondents reported increased productivity when using three or fewer AI tools, but self-reported productivity plummeted when respondents used four or more tools, with workers saying they felt brain fog or made more small mistakes as a result of technology overuse.

Nickle LaMoreaux, IBM's chief human resources officer, said this year the tech giant [would triple its number of young hires](#), suggesting that despite AI's ability to automate some of the required tasks, displacing entry-level workers would create a dearth of middle managers down the line, endangering the company's leadership pipeline.

What could reverse AI's productivity pattern?

To be sure, this productivity pattern could reverse. The IT boom of the 1970s and '80s eventually gave way to a surge of productivity in the 1990s and early 2000s, including a [1.5% increase in productivity growth](#) from 1995 to 2005 following decades of slump.

Economist and Stanford University's Digital Economy Lab director Erik Brynjolfsson noted in a [Financial Times op-ed](#) the trend [may already be reversing](#). He observed that fourth-quarter GDP was tracking up 3.7%, despite last week's jobs report revising down job gains to just 181,000, suggesting a productivity surge. His own analysis indicated a U.S. productivity jump of 2.7% last year, which he attributed to a transition from AI investment to reaping the benefits of the technology. Former Pimco CEO and economist Mohamed El-Erian also noted [job growth and GDP growth continuing to decouple](#) as a result in part of continued AI adoption, a similar phenomenon that occurred in the 1990s with office automation.

Some productivity increases may be hiding in plain sight. A [study](#) led by the Stanford Institute for Economic Policy Research found using internet browsing data from 200,000 U.S. households that generative AI increased the efficiency of online tasks like job hunting, travel planning, or shopping from between 76% and 176%. However,

researchers found the time AI users saved on chores was spent hanging out with friends or watching television, as opposed to spent on work on new skills development.

Slok saw the future impact of AI as potentially resembling a “J-curve” of an initial slowdown in performance and results, followed by an exponential surge. He said whether AI’s productivity gains would follow this pattern would depend on the value created by AI.

So far, AI’s path has already diverged from its IT predecessor. Slok noted in the 1980s, an innovator in the IT space had monopoly pricing power until competitors could create similar products. Today, however, AI tools are readily accessible as a result of “fierce competition” between large language model-buildings driving down prices.

Therefore, Slok posited, the future of AI productivity would depend on companies’ interest in taking advantage of the technology and continuing to incorporate it into their workplaces. “In other words, from a macro perspective, the value creation is not the product,” Slok said, “but how generative AI is used and implemented in different sectors in the economy.”

A version of this story was published on Fortune.com on Feb. 17, 2026.

More on AI and productivity:

- AI is [making productivity obsolete](#). The leaders who thrive next will have something machines can’t touch
- AI promised supreme productivity, but it’s [actually straining workloads](#) for employees—time spent emailing has doubled, and focused work sessions fell by 9%
- AI promises to free workers from grunt work, but psychologists say those mindless tasks are [exactly what our brains](#) need to recover

In 2001, Fortune first convened “The Smartest People We Know,” bringing together CEOs and founders, builders and investors, thinkers and doers. Since then, **Fortune Brainstorm Tech** has been the place where bold ideas collide. From **June 8–10**, we will return to **Aspen**—where it all began—to mark 25 years of Brainstorm. [Register now.](#)

About the Author



By **Sasha Rogelberg**
Reporter



Sasha Rogelberg is a reporter and former editorial fellow on the news desk at *Fortune*, covering retail and the intersection of business and popular culture.

[See full bio >](#)

Latest in AI

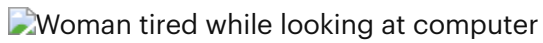
 Elon Musk says saving for retirement is irrelevant because AI is going to create a world of abundance. 'It won't matter' 6 hours ago

FUTURE OF WORK • ELON MUSK

Elon Musk says saving for retirement is irrelevant because AI is going to create a world of abundance: 'It won't matter'

By **Marco Quiroz-Gutierrez**

April 26, 2026



COMMENTARY • PRODUCTIVITY

AI is frying our brains — here's what leaders need to do about It

By **David Rock** and **Chris Weller**

April 26, 2026

BIG TECH • MICROSOFT

More than 90,000 tech workers have been laid off this year. But here's why companies like Microsoft are offering voluntary buyouts instead

By **Jacqueline Munis**

April 26, 2026

FUTURE OF WORK • GEN Z

‘You feel radicalized’: A Meta AI exec watched agents beat her top workers. Now she’s built a nonprofit to help Gen Z find jobs before they disappear

By **Jake Angelo**

April 26, 2026

 A Mark Cuban-backed AI startup is helping families turn conversations with their elderly relatives into lasting memories 1 day ago

STARTUPS & VENTURE • TECH

A Mark Cuban-backed AI startup is helping families turn conversations with their elderly relatives into lasting memories

By **Marco Quiroz-Gutierrez**

April 25, 2026

 Musk drops fraud claims against OpenAI, Altman ahead of trial

1 day ago

LAW • ELON MUSK

Musk drops fraud claims against OpenAI, Altman ahead of trial

By Madlin Mekelburg, Robert Burnson and Bloomberg

April 25, 2026

Most Popular

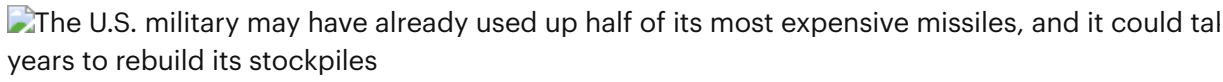
 **'You feel radicalized': A Meta AI exec watched agents beat her top workers. Now she's built a nonprofit to help Gen Z find jobs before they disappear** 16 hours ago

FUTURE OF WORK

'You feel radicalized': A Meta AI exec watched agents beat her top workers. Now she's built a nonprofit to help Gen Z find jobs before they disappear

By **Jake Angelo**

April 26, 2026

The U.S. military may have already used up half of its most expensive missiles, and it could take up to 4 years to rebuild its stockpiles 2 days ago

POLITICS

The U.S. military may have already used up half of its most expensive missiles, and it could take up to 4 years to rebuild its stockpiles

By **Sasha Rogelberg**

April 24, 2026

 Tim Cook built Apple into a \$4 trillion company. Then his greatest strength became his biggest 1 day ago

COMMENTARY

Tim Cook built Apple into a \$4 trillion company. Then his greatest strength became his biggest liability

By **Andrea Petrone**

April 25, 2026

 Despite nearing their 60s, nearly four in 10 Americans heading towards the end of their career have a retirement account 3 days ago 1

SUCCESS

Despite nearing their 60s, nearly four in 10 Americans heading towards the end of their careers don't even have a retirement account

By **Emma Burleigh**

April 23, 2026

 This CEO lived on canned soup and took just two days off for his daughter's birth. Now he admits he lost sight of proper work-life balance 2 days ago

SUCCESS

This CEO lived on canned soup and took just two days off for his daughter's birth. Now he admits he lost sight of proper work-life balance

By **Preston Fore**

April 25, 2026

 More than 90,000 tech workers have been laid off this year. But here's why companies like Microsoft are offering voluntary buyouts instead 12 hours ago

BIG TECH

More than 90,000 tech workers have been laid off this year. But here's why companies like Microsoft are offering voluntary buyouts instead

By **Jacqueline Munis**
April 26, 2026



Rankings ▼

Sections ▼

Customer Support ▼

Commercial Services ▼

About Us

Lists Calendar

[Press Center](#)

[Work At Fortune](#)

[Terms And Conditions](#)

[Site Map](#)



© 2026 Fortune Media IP Limited. All Rights Reserved. Use of this site constitutes acceptance of our [Terms of Use](#) and [Privacy Policy](#) | [CA Notice at Collection and Privacy Notice](#) | [Do Not Sell/Share My Personal Information](#)

FORTUNE is a trademark of Fortune Media IP Limited, registered in the U.S. and other countries. FORTUNE may receive compensation for some links to products and services on this website. Offers may be subject to change without notice.